

2VG3 Homework 4 - Collision Results

Generated from `rigidbody2d\_collision.py`.

Coefficient of Restitution $e = 1$

Scenario 0

Default: no initial rotation.

Collision events:

- Event 1: $t=2.233000$ s, $\text{normal}=(1.000000, 0.000000, -0.000000)$, $\text{avg_contact}=(0.266500, 25.000000, 0.000000)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.266000, 25.000000, -1.300000)$, $(0.267000, 25.000000, 1.300000)$
- Corner body: undetermined; Face body: undetermined. Note: contact geometry is symmetric.
- Event 2: $t=2.492000$ s, $\text{normal}=(0.661757, 0.000000, 0.749718)$, $\text{avg_contact}=(0.007500, 25.000000, 0.000000)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.394639, 25.000000, -0.342105)$, $(-0.379639, 25.000000, 0.342105)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{\text{trans}}=10.00000000$, $K_{\text{rot}}=0.00000000$, $K_{\text{total}}=10.00000000$
- After: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{\text{trans}}=2.85950413$, $K_{\text{rot}}=7.14049587$, $K_{\text{total}}=10.00000000$

Energy discussion:

- With $e=1$, total kinetic energy remains nearly unchanged; translational and rotational parts trade with each other via off-centre impulse.

Scenario 1

Body 0 starts at 30 degrees.

Collision events:

- Event 1: $t=2.123500$ s, $\text{normal}=(1.000000, 0.000000, -0.000000)$, $\text{avg_contact}=(0.704000, 25.000000, -1.157203)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.704000, 25.000000, -1.160489)$, $(0.704000, 25.000000, -1.153917)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.
- Event 2: $t=2.257500$ s, $\text{normal}=(0.958427, 0.000000, 0.285339)$, $\text{avg_contact}=(-0.288900, 25.000000, 0.991331)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(-0.288320, 25.000000, 0.989382)$, $(-0.289480, 25.000000, 0.993279)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{trans}=10.00000000$, $K_{rot}=0.00000000$, $K_{total}=10.00000000$

- After: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{trans}=2.50499225$, $K_{rot}=7.49500775$, $K_{total}=10.00000000$

Energy discussion:

- With $e=1$, total kinetic energy remains nearly unchanged; translational and rotational parts trade with each other via off-centre impulse.

Scenario 2

Body 0 starts at 60 degrees; body 1 starts at 30 degrees.

Collision events:

- Event 1: $t=2.109500$ s, $normal=(0.866025, 0.000000, 0.500000)$, $avg_contact=(0.678505, 25.000000, 0.158846)$, $type=averaged$.

- Raw contacts: $(0.678675, 25.000000, 0.158550)$, $(0.678334, 25.000000, 0.159141)$

- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

- Event 2: $t=2.433000$ s, $normal=(0.999453, 0.000000, 0.033082)$, $avg_contact=(-0.262000, 25.000000, 0.225449)$, $type=averaged$.

- Raw contacts: $(-0.261881, 25.000000, 0.221860)$, $(-0.262119, 25.000000, 0.229038)$

- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{trans}=10.00000000$, $K_{rot}=0.00000000$, $K_{total}=10.00000000$

- After: $P=(-2.00000000, 0.00000000)$, $L=3.00029321$, $K_{trans}=3.58035195$, $K_{rot}=6.41964805$, $K_{total}=10.00000000$

Energy discussion:

- With $e=1$, total kinetic energy remains nearly unchanged; translational and rotational parts trade with each other via off-centre impulse.

Scenario 3

Body 0 starts with $\omega=-5$ rad/s; body 1 is moved to $(-1.9, 25, 1.9)$ and starts at rest.

Collision events:

- Event 1: $t=0.051000$ s, $normal=(1.000000, 0.000000, -0.000000)$, $avg_contact=(-3.700000, 25.000000, 0.778640)$, $type=averaged$.

- Raw contacts: $(-3.700000, 25.000000, 0.774481)$, $(-3.700000, 25.000000, 0.782798)$

- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(2.00000000, 0.00000000)$, $L=-2.33333333$, $K_{trans}=2.00000000$, $K_{rot}=8.33333333$, $K_{total}=10.33333333$

- After: $P=(2.00000000, 0.00000000)$, $L=-2.33333333$, $K_{trans}=3.71689629$, $K_{rot}=6.61643704$, $K_{total}=10.33333333$

Energy discussion:

- With $e=1$, total kinetic energy remains nearly unchanged; translational and rotational parts trade with each other via off-centre impulse.

Scenario 4

Body 0 starts with $\omega=+3$ rad/s; body 1 starts at -10 degrees.

Collision events:

- Event 1: $t=2.163000$ s, $\text{normal}=(0.978587, 0.000000, 0.205833)$, $\text{avg_contact}=(0.262558, 25.000000, -0.957227)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.263251, 25.000000, -0.960523)$, $(0.261865, 25.000000, -0.953931)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.
- Event 2: $t=2.284000$ s, $\text{normal}=(0.881251, 0.000000, 0.472649)$, $\text{avg_contact}=(-0.077460, 25.000000, 1.215218)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(-0.076809, 25.000000, 1.214005)$, $(-0.078111, 25.000000, 1.216431)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=5.00000000$, $K_{\text{trans}}=10.00000000$, $K_{\text{rot}}=3.00000000$, $K_{\text{total}}=13.00000000$
- After: $P=(-2.00000000, 0.00000000)$, $L=5.00137667$, $K_{\text{trans}}=1.94170055$, $K_{\text{rot}}=11.05829945$, $K_{\text{total}}=13.00000000$

Energy discussion:

- With $e=1$, total kinetic energy remains nearly unchanged; translational and rotational parts trade with each other via off-centre impulse.

Coefficient of Restitution $e = 0$

Scenario 0

Default: no initial rotation.

Collision events:

- Event 1: $t=2.233000$ s, $\text{normal}=(1.000000, 0.000000, -0.000000)$, $\text{avg_contact}=(0.266500, 25.000000, 0.000000)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.266000, 25.000000, -1.300000)$, $(0.267000, 25.000000, 1.300000)$
- Corner body: undetermined; Face body: undetermined. Note: contact geometry is symmetric.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{\text{trans}}=10.00000000$, $K_{\text{rot}}=0.00000000$, $K_{\text{total}}=10.00000000$
- After: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{\text{trans}}=1.66942149$, $K_{\text{rot}}=1.78512397$, $K_{\text{total}}=3.45454545$

Energy discussion:

- With $e=0$, relative normal motion is damped at impact, so translational kinetic energy drops and some energy appears as rotation; total kinetic energy decreases.

Scenario 1

Body 0 starts at 30 degrees.

Collision events:

- Event 1: $t=2.123500$ s, $\text{normal}=(1.000000, 0.000000, -0.000000)$, $\text{avg_contact}=(0.704000, 25.000000, -1.157203)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.704000, 25.000000, -1.160489)$, $(0.704000, 25.000000, -1.153917)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.
- Event 2: $t=2.300500$ s, $\text{normal}=(0.932726, 0.000000, 0.360587)$, $\text{avg_contact}=(-0.369177, 25.000000, 1.421141)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(-0.345916, 25.000000, 1.360972)$, $(-0.392438, 25.000000, 1.481310)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.
- Event 3: $t=2.316500$ s, $\text{normal}=(0.926541, 0.000000, 0.376193)$, $\text{avg_contact}=(0.461166, 25.000000, -0.685031)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.789964, 25.000000, -1.494842)$, $(0.132368, 25.000000, 0.124780)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.
- Event 4: $t=2.320000$ s, $\text{normal}=(0.926187, 0.000000, 0.377065)$, $\text{avg_contact}=(-0.288505, 25.000000, 1.148798)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(-0.147125, 25.000000, 0.801526)$, $(-0.429885, 25.000000, 1.496071)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.
- Event 5: $t=2.333500$ s, $\text{normal}=(0.925331, 0.000000, 0.379161)$, $\text{avg_contact}=(0.702059, 25.000000, -1.308129)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.781316, 25.000000, -1.501554)$, $(0.622802, 25.000000, -1.114703)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{\text{trans}}=10.00000000$, $K_{\text{rot}}=0.00000000$, $K_{\text{total}}=10.00000000$
- After: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{\text{trans}}=5.46641995$, $K_{\text{rot}}=1.87375194$, $K_{\text{total}}=7.34017188$

Energy discussion:

- With $e=0$, relative normal motion is damped at impact, so translational kinetic energy drops and some energy appears as rotation; total kinetic energy decreases.

Scenario 2

Body 0 starts at 60 degrees; body 1 starts at 30 degrees.

Collision events:

- Event 1: $t=2.109500$ s, $\text{normal}=(0.866025, 0.000000, 0.500000)$, $\text{avg_contact}=(0.678505, 25.000000, 0.158846)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.678675, 25.000000, 0.158550)$, $(0.678334, 25.000000, 0.159141)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

- Event 2: $t=2.520000$ s, $\text{normal}=(0.999888, 0.000000, 0.014964)$, $\text{avg_contact}=(-0.062302, 25.000000, -0.744033)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(-0.062276, 25.000000, -0.745769)$, $(-0.062328, 25.000000, -0.742296)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=3.00000000$, $K_{\text{trans}}=10.00000000$, $K_{\text{rot}}=0.00000000$, $K_{\text{total}}=10.00000000$
- After: $P=(-2.00000000, 0.00000000)$, $L=3.00029321$, $K_{\text{trans}}=4.27344958$, $K_{\text{rot}}=1.60491201$, $K_{\text{total}}=5.87836159$

Energy discussion:

- With $e=0$, relative normal motion is damped at impact, so translational kinetic energy drops and some energy appears as rotation; total kinetic energy decreases.

Scenario 3

Body 0 starts with $\omega=-5$ rad/s; body 1 is moved to $(-1.9, 25, 1.9)$ and starts at rest.

Collision events:

- Event 1: $t=0.051000$ s, $\text{normal}=(1.000000, 0.000000, -0.000000)$, $\text{avg_contact}=(-3.700000, 25.000000, 0.778640)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(-3.700000, 25.000000, 0.774481)$, $(-3.700000, 25.000000, 0.782798)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(2.00000000, 0.00000000)$, $L=-2.33333333$, $K_{\text{trans}}=2.00000000$, $K_{\text{rot}}=8.33333333$, $K_{\text{total}}=10.33333333$
- After: $P=(2.00000000, 0.00000000)$, $L=-2.33333333$, $K_{\text{trans}}=1.10507356$, $K_{\text{rot}}=3.67133132$, $K_{\text{total}}=4.77640488$

Energy discussion:

- With $e=0$, relative normal motion is damped at impact, so translational kinetic energy drops and some energy appears as rotation; total kinetic energy decreases.

Scenario 4

Body 0 starts with $\omega=+3$ rad/s; body 1 starts at -10 degrees.

Collision events:

- Event 1: $t=2.163000$ s, $\text{normal}=(0.978587, 0.000000, 0.205833)$, $\text{avg_contact}=(0.262558, 25.000000, -0.957227)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(0.263251, 25.000000, -0.960523)$, $(0.261865, 25.000000, -0.953931)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.
- Event 2: $t=2.373500$ s, $\text{normal}=(0.940459, 0.000000, 0.339907)$, $\text{avg_contact}=(-0.477915, 25.000000, 1.520816)$, $\text{type}=\text{averaged}$.
- Raw contacts: $(-0.476695, 25.000000, 1.517442)$, $(-0.479134, 25.000000, 1.524190)$
- Corner body: both; Face body: both. Note: face-face simultaneous contacts.

First-contact conservation numbers:

- Before: $P=(-2.00000000, 0.00000000)$, $L=5.00000000$, $K_{trans}=10.00000000$, $K_{rot}=3.00000000$, $K_{total}=13.00000000$

- After: $P=(-2.00000000, 0.00000000)$, $L=5.00137667$, $K_{trans}=2.57722752$, $K_{rot}=2.67729775$, $K_{total}=5.25452527$

Energy discussion:

- With $e=0$, relative normal motion is damped at impact, so translational kinetic energy drops and some energy appears as rotation; total kinetic energy decreases.